

Karthik Katragadda

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PROFESSIONAL SUMMARY

DevOps Engineer with around **5** years of experience in automating and managing scalable AWS infrastructure within production environments. Proficient in enhancing CI/CD pipelines using **Terraform**, **Kubernetes**, **Jenkins**, and **GitOps**, resulting in faster, more reliable deployments. Experienced in implementing cloud security best practices and cost optimization through automation of IAM, **GuardDuty**, and auto-scaling, while ensuring compliance with **HIPAA** and SOC 2 standards. Seeking an AWS DevOps Engineer position to contribute to infrastructure **automation** and drive operational excellence in a high-scale environment.

TECHNICAL SKILLS

Scripting Languages	: Python, Java, JavaScript, SQL, Golang, Ruby, TypeScript, Bash, PowerShell, Groovy
Cloud Platforms	: AWS (IAM, S3, EC2, ECS, Far gate, RDS, Lambda, VPC, Code Pipeline, CloudFormation), Azure, OpenShift, VMware, GCP
IaC, DevOps Tools	: Terraform, Docker, Kubernetes (K8s, EKS, GKE, AKS), Ansible, Nexus, Packer, JSON, Chef, Puppet, YAML Build, CI/CD,
Version control	: Git, Maven, , GitOps, Jenkins, GitLab CI, GitHub Actions, Bitbucket, , Azure Bicep
Monitoring & Collaboration:	Grafana, Datadog, Cloud Watch logs, Confluence, New Relic, Splunk, ELK Stack, Prometheus,
Data Systems & Processing	: MySQL, No SQL, DynamoDB, MongoDB, PostgreSQL, Cassandra, RDBMS, Apache Kafka, Microsoft SQL Server, Redis
Web Servers & Networking	: Tomcat, RabbitMQ, TCP/IP, DNS, HTTP, HTTPS, RESTFUL API's, API Gateways, NGNIX, XML, Load balancing, Firewalls
Security & Compliance	: AWS (GuardDuty, Security Hub, Shield), SSL/TLS, RBAC, Hashi Corp Vault, FedRAMP, SOC 2, HIPAA, SAT/DAST, SIEM
Operating Systems, IDE	: Linux, Unix, Windows, Red Hat Enterprise Linux, CentOS, Ubuntu, Windows Server, Debian, Visual Studio Code

PROFESSIONAL EXPERIENCE

- AWS DevOps Engineer | Client: Deloitte | Austin, Texas**

June 2023 – Present
- Provisioned and automated **AWS** infrastructure on Linux systems using **Terraform**, **CloudFormation**, and **Ansible AWX**, leading to a 50% reduction in deployment errors and enabling high-availability **GPU**-powered **EC2** instances for **machine learning** inference workloads with 99.9% uptime.
 - Experienced DevOps Engineer** with strong expertise in implementing and managing **ArgoCD GitOps solutions** for automated deployments, configuration management, and application lifecycle.
 - Skilled in **designing, integrating, and supporting GitOps workflows**, ensuring seamless CI/CD pipelines and high developer productivity.
 - Developed full-cycle automation scripts using **Python**, **Golang**, **PowerShell**, and **JavaScript** to enhance provisioning, **CI/CD** integration, and API network diagnostics (traceroute, dig) with **YAML/JSON** handling—boosting team efficiency by 40%.
 - Instrumented observability using **CloudWatch Logs & Metrics**, **Datadog**, and **Prometheus/Grafana**, improving incident triage by 35% and enabling **performance monitoring** of key services, including **ML model-serving endpoints**.
 - Managed deployment of containerized microservices using **Docker**, **Amazon EKS**, **Helm**, and **Argo CD** with **NGINX ingress**, implementing Blue/Green release strategies that minimized production rollout failures by 60% for high-impact healthcare APIs.
 - Hardened infrastructure security and disaster recovery across **IaaS**, **PaaS**, and **SaaS** environments using **IAM**, **Security Groups**, **AWS WAF**, **Secrets Manager**, **TLS/SSL**, **PKI**, and **Vault**, ensuring **SOC 2**, **HIPAA**, and **FedRAMP** compliance across production systems.
 - Reduced cloud spends by 25% through **EC2 instance right-sizing**, **Auto Scaling Group tuning**, and **S3 lifecycle policies**, including optimization of **GPU node utilization** for **ML inference workloads**.
 - Executed **SRE best practices** by defining **SLIs**, **SLOs**, and **SLAs**, running **JUnit tests**, **post-incident reviews**, and collaborating across **Dev**, **QA**, and **IT** using **JIRA** and **Agile Scrum**, boosting deployment reliability by 25%.
- Azure DevOps Engineer | Wipro | Bengaluru, India**

Jun 2021 – May 2022
- Architected **auto-scaling strategies on AKS** using **HPA**, **Cluster AutoScaler**, and **custom metrics**, reducing 5xx errors by 65% during peak user loads and **GPU-backed inference service spikes** across pricing and recommendation APIs.
 - Streamlined **CI/CD pipelines** with **Azure DevOps**, **YAML**, and **Git Repos**, reducing deployment time by 50% and integrating automated rollout of **containerized model-serving endpoints** into **e-commerce workflows**.
 - Orchestrated **Azure IaaS and PaaS resources** using **Terraform**, **ARM templates**, and **Bicep**, provisioning **N-series GPU VMs** and codifying deployments for **ML inference workloads**, reducing environment setup and deployment time by 60%.
 - Established **observability** using **Azure Monitor**, **Log Analytics**, **Prometheus**, and **App Insights**, **cutting MTTD** by 45% and monitoring latency, availability, and error rates for **ML-powered APIs**.
 - Automated **AKS health checks**, **load tests**, and **deployment probes** using **Python**, **PowerShell**, and **Bash**, improving operational response by 40% and expanding day-2 automation coverage by 50%.

- Designed and implemented end-to-end CI/CD pipelines using **Jenkins**, **Bitbucket**, and **GitOps**, reducing deployment cycles by 60% for a major Capgemini financial services client, ensuring continuous delivery under stringent compliance frameworks.
- Automated provisioning of secure and scalable **AWS infrastructure** (EC2, RDS, S3) using **Terraform**, **Ansible**, and **CloudFormation**, resulting in a 40% improvement in uptime during peak transaction periods.
- Integrated **DevSecOps practices** into deployment workflows by incorporating **IAM**, **GuardDuty**, and **KMS**, achieving 99.9% security compliance and enabling real-time remediation of critical vulnerabilities.
- Deployed and managed **containerized microservices** using **Docker**, **Kubernetes**, and **Helm** on **AWS EKS**, increasing availability of payment and fraud analytics APIs by 35% during high-load events.
- Built custom automation and monitoring tools using **Python**, **Bash**, and **PowerShell**, cutting Mean Time to Detect (MTTD) by 45% and streamlining operational workflows across cloud environments.
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EDUCATION

Lamar University <i>Master of Science, Computer Science</i>	May 2024
KLU University <i>Bachelor of Engineering, Computer Science</i>	April 2021